EPAT Structure

Modules	# Lectures*	Suggested Hours†
PRIMER	NA	35
Statistics for Financial Markets	3	20
Python: Basics & Its Quant Ecosystem	2	15
Market Microstructure for Trading	4	24
Equity, FX & Futures Strategies	5	48
Data Analysis & Modeling in Python	5	30
Machine Learning for Trading	7	48
Trading Tech, Infra & Operations	2	10
Advanced Statistics for Quant Strategies	2	15
Trading & Back-testing Platforms	5	36
Portfolio Optimization & Risk Management	2	12
Options Trading & Strategies	5	35
EXAM	NA	NA

* The number of lectures mentioned is tentative. You will be notified of the scheduled lecture a week in advance. The suggested hours include self-study hours in addition to the lectures of about 3 hours each.

Optional Sessions

- Two preparatory sessions will be conducted before the Orientation session, one doubt-solving session on Statistics Primer and one doubt-solving session on Python Primer.
- A total of three Python tutorials will be conducted after four Python lectures to answer queries and resolve doubts about Python.

Primer

The EPAT Primer is a collection of material we have put together on the following topics.

- Stock Market Basics

- Excel
- Statistics
- Python
- Options
- Machine Learning
- Macroeconomics

Our aim in creating this is to introduce the fundamentals of specific vital ideas to students who are unfamiliar with them. From our past experience, we have seen that students have varying degrees of understanding of the topics mentioned above. We, therefore, recommend that you devote time to each of them accordingly (i.e., the newer the topic is to you, the higher proportionate time you spend on it). These are the minimum required readings for you before getting started with the EPAT lectures.

The objectives are:

- To learn about stock markets and get a brief understanding of how they work. You will learn about several concepts like financial instruments, intermediaries, types of traders, investors, and their roles.

- Starting from the very basics, moving on to available functions used with lots of examples to give you full clarity and understanding of Excel.

- To understand and implement basic statistics through examples. Topics covered include
    a. Data and Statistics, Difference Between Probability and Statistics, Statistics in Quant Finance & Algorithmic Trading
    b. Descriptive Statistics, Organizing and Presenting the Data, Data Types: Quantitative and Qualitative, Frequency-Distribution of Data
    c. Standard Statistical Distributions, Basics of Inferential Statistics, Covariance& Correlation, Hypothesis Testing

- Starting from installation to using basic functions in Python with exercises in the notebook for your complete understanding.

- To understand the basics of Options - associated terminology, options pricing basic with factors affecting option prices and their quantification (Greeks), simple option trading strategies.

- An introduction to Machine Learning for Trading covers:
    a. Supervised, Unsupervised and Reinforcement Machine Learning.
    b. Different types of Machine Learning algorithms such as Linear regression, Logistic regression, K - Nearest Neighbor, Artificial Neural Networks, Random Forest, and Support Vector Machine.
    c. Application of Machine Learning in trading.

Statistics for Financial Markets

The main tools for quantitative trading include Statistics and Excel. This module will take you through the application of these tools and help you appreciate their importance.

The learning objectives are:

- Basics of Excel, application of some trading strategies in Excel. Using Excel to create a back-testing model for a given hypothesis.

- The art of visualizing data. Statistics and probability concepts (Bayesian and Frequentist methodologies), moments of data and Central Limit Theorem.
- Applications of statistics- Random Walk Model for predicting future stock prices using simulations and understanding results, Capital Asset Pricing Model.
- Short Introduction to Modern Portfolio Theory - statistical approximations of risk/reward.

Python: Basics & Its Quant Ecosystem

One of the key things of Algorithmic Trading is back-testing your strategy, and you need a certain level of programming skills for the same. This module will help you understand the application of Python concepts required for writing and back-testing your trading strategies.

The learning objectives are:

- Understanding of data types, variables, Python in-built data structures, inbuilt functions, logical operators such as and, or, control structures: If, nested ifs, loops: for, and while.
- Writing Python functions to implement the strategy.
- Introduction to some key libraries NumPy, pandas, and Matplotlib.
- Installing and introduction to pandas-data reader.
- Writing and back-testing some basic strategies.

Market Microstructure for Trading

In this module, you will be introduced to Market Microstructure for Algorithmic Trading and Executing Strategies

The learning objectives are:

- Overview of Electronic and Algorithmic Trading.
- Various order types, order book dynamics, Spoofing, Price Time Priority Algorithm and Guerilla Algorithm.
- Execution strategy to trade large volumes.
- The algorithmic trading process from a market microstructure perspective.

Equity, FX & Futures Strategies

This module will take you through developing trading strategies across different asset classes.

The learning objectives are:

- Understanding of Equities Derivative markets, various parameters like OI, volume etc., and the impact of market players on the derivative parameters.

- VWAP strategy how to implement, effect of VWAP, how to maintain log journal.

- Basic optimization for various strategy parameters

  a) Pair Trading Methodology – Identification of probable pairs, entry-exit, and back-testing
  b) In-depth understanding of Stationarity & co-integration
  c) Position sizing and risk management

- Exposure to different types of Momentum (Time series momentum & Cross-sectional momentum).

- Trend strategy and Pair Trading strategy modeling in detail.

- ETF's trading, including arbitrage, market making, and asset allocation strategies using ETFs.

Data Analysis & Modeling in Python

The module aims to help you validate and backtest trading ideas using python

The learning objectives are:

- Learn to back-test a strategy in Python and measure the performance using Log Returns.

- Understand the basics of Object-Oriented Programming (OOP), programming layouts of various packages, and appreciate the classes over procedural programming style.

- Implement various OOP concepts in your Python program - Aggregation, Inheritance, Composition, Encapsulation, and Polymorphism.

- Back-testing methodologies & techniques for Long-Only as well as Long-Short strategies

Machine Learning for Trading

This module aims to help you implement machine-learning concepts to build trading strategies

The learning objectives are:

- Decision Trees, Support Vector Machines, Neural Networks, Forward propagation, Backward propagation, Various neural network architectures

- Building a "Principal Component Analysis" manually, conducting a pairs-trading back-test using PCA, Simulation of multiple co-integrated assets, Sector statistical arbitrage using PCA

- Using Python and Jupyter Notebooks to create features, evaluate models, use feature selection, and test raw performance

- Overview of Alternate Data: Sources, data formats, storage and retrieval choices, Understanding RDF and Knowledge Graph, Tagging Unstructured Data with relevant metadata
- Using spaCy for common Text processing tasks, Understanding Topic Modeling, and Topic Classification
- Understanding Machine Readable News Programmatic consumption of news
- Machine Readable News in the Financial Industry: Sample in Production use cases, Sentiment Data in the Financial Industry: Sample in Production use cases
- Basic ideas of deep reinforcement learning such as reward, explore/exploit, Bellman equation, and memory replay.
- Challenges and problems with RL in trading, Implementation of RL in a simple strategy using "gamification".

Trading Tech, Infra & Operations

The learning objectives are:

- To understand in detail, the System Architecture of a traditional trading system and compare it to an automated trading system.
- To understand the needs, requirements, processes, advantages, and applications of Algorithmic Trading.
- Understanding the infrastructure (hardware, physical, network, etc.) requirements.
- Understanding the business environment (including the regulatory environment, financials, business insights, etc.) while setting up an Algorithmic Trading desk.

Advanced Statistics for Quant Strategies

This is an advanced module designed to take you through the application of Statistical parameters in Quantitative Trading Strategies.

The learning objectives are:

· Study time series analysis and statistical parameters such as autocorrelation function, partial autocorrelation function, maximum likelihood estimation, Akaike Information Criterion. Understanding what time series is and the terminology associated with it, learning common features of financial asset returns, Exponential smoothing, ARIMA (Auto Regressive Integrated Moving Average), GARCH (Generalized Auto Regressive Conditional Heteroskedasticity)

Trading & Back-testing Platforms

This module aims to help you understand the implementation of your strategies in the live trading environment and will introduce you to platforms like Interactive Brokers using Python programming language

The learning objectives are:

- Using IBridgePy API to automate your trading strategies on the Interactive Brokers platform. Strategy Life Cycle, Work with IB TWS, Understand IB TWS API Architecture, Explain building blocks of IB API, Overview of Cloud

Computing

- Understand how the REST API works, know how to work with web sockets in Python

- Components and steps to start an investment fund, Different Python solutions for algo trading, IBridgePy cornerstone functions, Improvements to moving average crossover strategy, Build a buy-low-sell-high strategy, Backtest strategies

- Strategy flow in Blueshift, Manage your portfolio in Blueshift, Demo strategies for back-testing

Portfolio Optimization & Risk Management

One of the important aspects of Algorithmic Trading is portfolio optimization and risk management. This module is dedicated to identifying and managing the risks involved in different kinds of strategies.

The learning objectives are:

- Different methodologies of evaluating portfolio & strategy performance.

- Know about Risk Management structures & policies, sources of risk, risk limits & risk components evaluation, risk control systems

- Understand risk and returns terminology, Conceptualize the portfolio building process, Build a portfolio with multiple stocks, Profitability analysis of a portfolio and strategy

Options Trading & Strategies

This module will take you through the world of trading in Options. Options trading strategies help you gain exposure to a specific type of opportunity or risk while attempting to mitigate other types of risks.

The learning objectives are:

- General option trading philosophy, Trading Principles: edge, variance, breadth, stops, History of derivatives, Pricing forwards and futures through the idea of no-arbitrage, Introduction to options, Payoff diagrams and common option structures

- Model-independent features of options: arbitrage relationships between various options, and options and underlying, Option pricing variables and parameters, A toy binomial pricing model, Risk-neutrality, Deriving the Black-Scholes-Merton PDE, Properties of BSM solution

- Some empirical properties of volatility, Volatility measurement and forecasting, Contextualizing forecasts, Implied volatility, The variance premium

- The variance premium, Covered calls, Using the term structure as a predictor, Equity options and earnings, Selecting strikes and expirations, Hedging in practice

- Expiration trading, Early exercise, Trade planning and evaluation, Risk measures, An example that summarizes the volatility trading process

EPAT Project (Optional)

- Write your working strategy starting from ideation, data analysis, strategy formulation, back-testing, and coding.

- Mentorship under a domain expert, or practitioner.

- Project topics include, but not limited to, Pair trading strategies, Dispersion Trading, Machine Learning, Skew Trading, Volatility Smile, Forward Volatility (You can check some of the past projects works at https://www.quantinsti.com/category/project-work-epat)

- Submission Requirements
    a) Working Strategy Model
    b) Project Report 2000-5000 words long

EPAT Mid-Term Exam

- The EPAT Mid-Term exam has a 10% weightage

- The Mid-Term exam is MCQ based test conducted on the modules covered in the first two months of the EPAT programme

- The duration of the exam will be 60 minutes for 30 MCQ questions

- Unlike the EPAT final exam, the Mid-Term exam is not proctored.

- You will have only one attempt to write the exam

EPAT Final Exam

- We offer the world's first verified certification in algorithmic and quantitative trading through proctored EPAT certification exams worldwide.

- The exams are conducted through the joint efforts of QuantInsti & our examination partner Prometric, one of the leading test administrators globally.

Duration: 4 hours (Includes reaching 30 minutes before the exam start time and the break of up to 30 mins between the two sections)

Format: Divided into two sections:

- EPAT Final Test contains MCQs (Multiple Choice Questions)

- EPAT Final Assignment contains Assignment type questions

- In case you finish the Quiz section, and there is spare time, the remaining time will be added to the Assignment section

- You may take a break of up to 30 minutes (not included in the duration of either of the sections) between the two sections. Please note that the second section will start automatically after 30 minutes of submitting the first section.